SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1021	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	12-10-2026	**Duration:**	60 minutes

Code of Conduct

I P.Vishwa nandan *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Vishwa nandan

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

Industry Context:

Deep-sea mining is an emerging field aimed at collecting valuable battery metals (like cobalt and nickel) from the ocean floor. However, scraping the seabed stirs up massive clouds of sediment (turbidity plumes) that can suffocate marine life if they drift outside the mining zone. Environmental protection rules require mining companies to track these sediment plumes continuously and stop operations immediately if they cross strict safety boundaries.

Problem Statement:

Design and develop a fault-tolerant Deep-Sea Mining Environmental Compliance System that tracks underwater sediment plumes and enforces environmental boundaries. The system architecture must coordinate fleets of surface buoys and autonomous underwater drones. It must ingest real-time data streams—including water turbidity, current direction variables, acoustic sound profiles, and tool depth coordinates—to manage compliance checks.

1. Identify the Project Vision and Stakeholders

1.1 Project Vision

The vision of this project is to develop a fault-tolerant environmental monitoring system that continuously tracks sediment plumes generated during deep-sea mining. The system coordinates surface buoys and autonomous underwater drones to collect real-time environmental data, detect safety boundary violations, generate alerts, and initiate mining shutdown procedures when required.

The system aims to protect marine ecosystems, improve environmental compliance, reduce monitoring delays, and provide reliable environmental records even when individual sensors or communication links fail.

1.2 Project Objectives

The main objectives of the proposed system are:

1. To collect real-time environmental data from surface buoys and autonomous underwater drones.
2. To monitor water turbidity, current direction and speed, acoustic sound profiles, and mining tool depth coordinates.
3. To validate sensor data and identify missing, outdated, or incorrect measurements.
4. To analyze sediment plume movement and detect environmental boundary violations.
5. To generate alerts when environmental safety limits are approached or exceeded.
6. To initiate the required mining shutdown procedure when an approved safety condition is violated.
7. To detect sensor failures and communication interruptions.
8. To maintain environmental records and generate compliance reports.
9. To provide secure access to environmental information through a user-friendly dashboard.
10. To improve the reliability and safety of deep-sea mining operations.

1.3 Key Stakeholders

1. Mining Company

- Provides the resources and infrastructure required for the project.
- Operates the mining equipment.
- Ensures compliance with environmental requirements.
- Maintains safe mining operations.

2. Mining Operators

- Monitor environmental conditions through the dashboard.
- Review alerts generated by the system.

- Supervise mining activities.
- Respond to environmental warnings and shutdown notifications.

3. Environmental Regulators

- Define or approve environmental safety limits and geographical boundaries.
- Monitor compliance with applicable environmental regulations.
- Review environmental monitoring reports.
- Investigate reported environmental violations.

4. Marine Scientists

- Study sediment plume movement and marine ecosystems.
- Analyze turbidity, current measurements, and acoustic information.
- Evaluate potential environmental impacts.
- Help define scientifically appropriate monitoring rules.

5. Surface Buoy Operators

- Maintain surface buoys and their sensors.
- Monitor communication equipment.
- Ensure that environmental data are transmitted correctly.
- Report equipment failures.

6. Autonomous Drone Engineers

- Develop and maintain underwater drones.
- Integrate environmental sensors into the drones.
- Maintain navigation and communication systems.
- Ensure that drones collect accurate measurements.

7. Product Owner

- Represents stakeholder needs and expectations.
- Defines product priorities.
- Maintains and prioritizes the product backlog.
- Reviews completed features and provides feedback to the development team.

8. Scrum Master

- Facilitates Scrum meetings and sprint planning.
- Helps the team follow Agile practices.
- Identifies and helps remove obstacles.
- Supports communication between team members and stakeholders.

9. Development Team

- Designs and implements software modules.
- Integrates environmental sensors and monitoring devices.
- Develops the dashboard, data processing, and compliance features.
- Maintains technical documentation.

10. Test Engineers

- Prepare and execute test cases.
- Verify functional requirements and acceptance criteria.
- Test sensor failures, communication interruptions, and recovery.
- Identify defects and verify their resolution.

11. Data Engineers

- Manage environmental data collection and storage.
- Validate sensor readings.
- Process real-time data streams.
- Maintain data quality and synchronization.

12. Maintenance Team

- Inspects and repairs sensors, drones, and buoys.
- Replaces faulty equipment.
- Monitors equipment health.
- Supports system recovery after failures.

13. Environmental Auditors

- Review environmental records and reports.
- Verify compliance with monitoring requirements.
- Examine alerts and shutdown events.
- Check whether required environmental procedures were followed.

14. Cybersecurity Specialists

- Protect the system against unauthorized access.
- Implement authentication and access control.
- Secure environmental data and communication.
- Monitor cybersecurity risks.

1.4 Selected Agile Framework: Scrum

The Scrum framework is selected for developing the Deep-Sea Mining Environmental Compliance System because it supports iterative development, continuous feedback, frequent testing, and flexible prioritization.

The project is divided into short development cycles called sprints. Each sprint focuses on a specific set of requirements and produces a working increment of the system.

A two-week sprint duration is proposed for this project.

The initial sprint will focus on device registration, simulated environmental data collection, data validation, basic device-health monitoring, and dashboard development. Later sprints will implement environmental analysis, boundary detection, fault recovery, alerts, and shutdown integration.

Because this is a safety-critical system, every sprint must include appropriate testing and requirements traceability. Changes affecting shutdown behavior require additional safety review before deployment.

2. Create Epics and User Stories

2.1 Definition of Epics

An epic is a large requirement that can be divided into smaller user stories.

The following epics are identified for this project.

Epic E1: Device Management

This epic covers the registration and management of surface buoys and autonomous underwater drones.

Epic E2: Sensor Data Collection

This epic focuses on collecting environmental measurements from the supported monitoring devices.

Epic E3: Data Validation

This epic covers checking environmental measurements for correctness, completeness, freshness, and consistency.

Epic E4: Environmental Visualization

This epic provides a dashboard for displaying sensor readings, device status, and environmental conditions.

Epic E5: Environmental Analysis

This epic focuses on analyzing sediment plume movement and identifying possible environmental risks.

Epic E6: Compliance Monitoring

This epic compares environmental measurements and estimated plume movement against approved safety boundaries.

Epic E7: Alert and Shutdown Management

This epic handles environmental alerts and initiates the required shutdown procedure when approved safety conditions are violated.

Epic E8: Fault Detection and Recovery

This epic identifies sensor failures, communication interruptions, and software faults while supporting recovery.

Epic E9: Data Storage and Reporting

This epic stores environmental measurements, alerts, shutdown events, and compliance records.

Epic E10: Security and Access Control

This epic provides authentication, user permissions, and protection for sensitive environmental information.

2.2 User Stories

User stories describe system requirements from the perspective of a user and the benefit they expect.

The standard format is:

As a <user>, I want <feature>, so that <benefit>.

US-01: Register Monitoring Devices

Epic: Device Management

As a system administrator, I want to register surface buoys and underwater drones, so that the system can identify and manage each monitoring device.

US-02: Monitor Device Status

Epic: Device Management

As a system administrator, I want to view the operational status of each device, so that I can identify unavailable or faulty equipment.

US-03: Collect Environmental Data

Epic: Sensor Data Collection

As an environmental monitoring operator, I want the system to collect real-time turbidity, current, acoustic, and depth measurements, so that I can monitor underwater conditions.

US-04: Validate Sensor Readings

Epic: Data Validation

As a data engineer, I want the system to validate sensor readings, so that incorrect or incomplete data do not lead to unsafe environmental decisions.

US-05: View Environmental Conditions

Epic: Environmental Visualization

As a mining operator, I want to view environmental measurements and device locations on a dashboard, so that I can understand the current environmental situation.

US-06: Analyze Sediment Plume Movement

Epic: Environmental Analysis

As a marine scientist, I want the system to analyze turbidity and current measurements, so that I can estimate sediment plume movement and identify potential environmental risks.

US-07: Configure Environmental Boundaries

Epic: Compliance Monitoring

As an authorized environmental regulator, I want approved environmental safety limits and geographical boundaries to be configured in the system, so that mining activities can be evaluated against applicable requirements.

US-08: Detect Boundary Violations

Epic: Compliance Monitoring

As an environmental monitoring operator, I want the system to detect sediment plumes crossing approved safety boundaries, so that environmental violations can be identified promptly.

US-09: Generate Environmental Alerts

Epic: Alert and Shutdown Management

As a mining operator, I want to receive alerts when environmental safety limits are violated, so that appropriate safety actions can be initiated.

US-10: Initiate Mining Shutdown

Epic: Alert and Shutdown Management

As a mining safety officer, I want the system to initiate the approved shutdown procedure when a defined safety violation occurs, so that environmental damage can be minimized.

US-11: Detect System Failures

Epic: Fault Detection and Recovery

As a system administrator, I want the system to detect sensor failures and communication interruptions, so that monitoring problems can be identified quickly.

US-12: Recover Data After Communication Failure

Epic: Fault Detection and Recovery

As a data engineer, I want locally buffered sensor data to be synchronized after communication is restored, so that temporary connection failures do not unnecessarily cause permanent data loss.

US-13: Maintain Compliance Records

Epic: Data Storage and Reporting

As an environmental auditor, I want the system to store sensor readings, alerts, and shutdown events, so that environmental compliance can be verified.

US-14: Generate Compliance Reports

Epic: Data Storage and Reporting

As an environmental regulator, I want to generate reports for a selected time period, so that I can review environmental conditions and safety events.

US-15: Manage User Permissions

Epic: Security and Access Control

As a system administrator, I want to assign permissions based on user roles, so that only authorized users can access sensitive information and safety-related controls.

3. Define Acceptance Criteria

Acceptance criteria describe the conditions that must be satisfied before a user story can be accepted.

The Given–When–Then format is used to make the expected behavior clear and testable.

US-01: Register Monitoring Devices

Given an authorized administrator is logged into the system,

When the administrator enters a unique device ID, device type, and required device information,

Then the system should register the device and display its unique ID and initial status.

Additional acceptance criteria:

1. The system must reject duplicate device IDs.
2. Required fields must be validated before saving.
3. Unauthorized users must not be able to register devices.

US-02: Monitor Device Status

Given a monitoring device is registered,

When the system receives device-health information or detects a missed heartbeat,

Then the dashboard must display the corresponding device status.

Additional acceptance criteria:

1. The system must distinguish available, unavailable, and unknown status.
2. A missing heartbeat must be evaluated against a configured timeout.
3. A device failure must be recorded.

US-03: Collect Environmental Data

Given a registered device is connected to the monitoring system,

When it sends a supported sensor observation,

Then the system must receive the observation with its device ID, timestamp, measurement type, and value.

Additional acceptance criteria:

1. The system must support configured turbidity, current, acoustic, and depth data formats.

2. Unsupported data formats must be rejected and logged.
3. Communication failures must be reported.

US-04: Validate Sensor Readings

Given the system receives an environmental observation,

When the observation is checked against the configured schema, data-quality rules, and timestamp requirements,

Then the system must mark it as valid or invalid and record the reason for rejection when applicable.

Additional acceptance criteria:

1. Missing required fields must be detected.
2. Duplicate observations must be identified.
3. Stale readings must not be treated as current measurements.
4. Invalid data must not independently establish that mining conditions are safe.

US-05: View Environmental Conditions

Given the system has received environmental measurements,

When an authorized operator opens the dashboard,

Then the dashboard must display the latest available readings, device status, and data timestamps.

Additional acceptance criteria:

1. The dashboard must show missing or outdated data clearly.
2. Critical alerts must be distinguishable from ordinary status messages.
3. The dashboard must display information according to the user's permissions.

US-06: Analyze Sediment Plume Movement

Given valid turbidity, current, and relevant location data are available,

When the environmental analysis module processes the measurements,

Then the system must produce an estimated plume movement result with its data-quality status.

Additional acceptance criteria:

1. The analysis must identify when essential input data are missing.
2. The system must not present uncertain estimates as confirmed measurements.
3. The analysis output must include a timestamp.

US-07: Configure Environmental Boundaries

Given an authorized user has permission to manage environmental limits,

When the user enters approved thresholds and geographical boundaries,

Then the system must validate and save the configuration.

Additional acceptance criteria:

1. Invalid configurations must be rejected.
2. Configuration changes must be recorded in an audit log.
3. Previous configurations must remain traceable.

US-08: Detect Boundary Violations

Given valid environmental measurements and an approved boundary configuration are available,

When the compliance engine determines that a defined violation condition has occurred,

Then the system must classify the condition as a violation and create a corresponding event.

Additional acceptance criteria:

1. The event must include a timestamp and relevant measurements.
2. Stale or insufficient data must be identified.
3. The detection result must be traceable to the applicable rule and configuration.

US-09: Generate Environmental Alerts

Given the system has detected a warning condition or confirmed safety violation,

When the alert rules are evaluated,

Then the system must generate an alert containing the event type, severity, timestamp, and required response.

Additional acceptance criteria:

1. Critical alerts must be visually distinguishable.
2. Alerts must be recorded.
3. The system must avoid generating uncontrolled duplicate alerts for the same event.

US-10: Initiate Mining Shutdown

Given a confirmed shutdown condition has been identified,

When the safety enforcement component processes the event,

Then the system must issue the configured shutdown request through the approved control interface.

Additional acceptance criteria:

1. The shutdown request must have a traceable event or command ID.
2. The system must record the response or missing acknowledgement.
3. A shutdown request must not be represented as a completed shutdown until the required confirmation is received.

4. The independent safety mechanism must remain available according to the approved safety design.

US-11: Detect System Failures

Given a registered device or software component is being monitored,

When a required health check fails or the configured timeout is exceeded,

Then the system must report the fault and update the relevant health status.

Additional acceptance criteria:

1. The fault must be time-stamped.
2. An important monitoring failure must generate an alert.
3. The approved fail-safe policy must be followed when essential monitoring is unavailable.

US-12: Recover Data After Communication Failure

Given a device has stored observations locally during a communication outage,

When the connection is restored,

Then the system must receive and reconcile the buffered observations.

Additional acceptance criteria:

1. Original timestamps must be preserved.
2. Duplicate observations must not be stored as separate unique events.
3. Missing data must be identified where recovery is impossible.

US-13: Maintain Compliance Records

Given sensor readings, alerts, or shutdown events are generated,

When the system processes these events,

Then it must store traceable records containing the relevant event information.

Additional acceptance criteria:

1. Records must contain timestamps and source identifiers where applicable.
2. Important records must be protected against unauthorized modification.
3. Authorized auditors must be able to retrieve records.

US-14: Generate Compliance Reports

Given environmental records exist for a selected time period,

When an authorized user requests a compliance report,

Then the system must produce a report containing the available environmental measurements, alerts, faults, and shutdown events for that period.

Additional acceptance criteria:

1. The selected reporting period must be displayed.
2. Missing data must be identified.
3. The report must include its generation timestamp.

US-15: Manage User Permissions

Given an administrator has permission to manage user accounts,

When the administrator assigns an approved role to a user,

Then the system must apply the associated permissions.

Additional acceptance criteria:

1. Unauthorized access attempts must be denied.
2. Permissions must be checked before sensitive operations.
3. Relevant access-control events must be recorded.

4. Prioritize the Product Backlog

4.1 Product Backlog Prioritization Method

The product backlog is prioritized according to environmental safety, business value, stakeholder needs, technical dependencies, risk reduction, and implementation feasibility.

The MoSCoW technique is used to classify requirements.

- **Must Have:** Essential functionality required for the system's intended purpose.
- **Should Have:** Important functionality that should be implemented soon.
- **Could Have:** Useful functionality that may be implemented when sufficient capacity is available.
- **Won't Have in This Release:** Functionality excluded from the current release but potentially considered in the future.

Story points are preliminary estimates using the Fibonacci sequence. These estimates should be reviewed by the development team during sprint planning.

4.2 Prioritized Product Backlog

Priority 1: US-01 — Register Monitoring Devices

Epic: Device Management

MoSCoW Priority: Must Have

Story Points: 3

This story allows the administrator to register surface buoys and underwater drones.

It is given the highest priority because the system must identify devices before it can reliably associate incoming sensor data with their sources.

Priority 2: US-03 — Collect Environmental Data

Epic: Sensor Data Collection

MoSCoW Priority: Must Have

Story Points: 5

This story enables the system to receive environmental readings from monitoring devices.

It is essential because environmental monitoring, analysis, and compliance checking depend on sensor data.

Priority 3: US-04 — Validate Sensor Readings

Epic: Data Validation

MoSCoW Priority: Must Have

Story Points: 3

This story checks environmental readings for missing fields, invalid values, duplicates, and stale timestamps.

It is important because incorrect or outdated measurements can affect environmental safety decisions.

Priority 4: US-02 — Monitor Device Status

Epic: Device Management

MoSCoW Priority: Must Have

Story Points: 3

This story allows administrators to monitor whether buoys, drones, and supported components are available.

It helps identify failed devices and missing observations that could affect monitoring reliability.

Priority 5: US-05 — View Environmental Conditions

Epic: Environmental Visualization

MoSCoW Priority: Must Have

Story Points: 5

This story provides a dashboard showing sensor readings, timestamps, device identifiers, and device-health status.

It is important because operators need to understand current environmental conditions and identify missing data.

Priority 6: US-07 — Configure Environmental Boundaries

Epic: Compliance Monitoring

MoSCoW Priority: Must Have

Story Points: 3

This story allows authorized users to configure approved environmental limits and geographical boundaries.

It must be implemented before meaningful boundary-compliance checks can be performed.

Priority 7: US-08 — Detect Boundary Violations

Epic: Compliance Monitoring

MoSCoW Priority: Must Have

Story Points: 5

This story enables the compliance engine to identify when a sediment plume or defined environmental condition violates approved safety requirements.

It is central to the environmental protection purpose of the system.

Priority 8: US-09 — Generate Environmental Alerts

Epic: Alert and Shutdown Management

MoSCoW Priority: Must Have

Story Points: 3

This story generates warnings and critical alerts when defined environmental conditions occur.

It helps operators respond promptly to environmental risks.

Priority 9: US-11 — Detect System Failures

Epic: Fault Detection and Recovery

MoSCoW Priority: Must Have

Story Points: 5

This story identifies sensor failures, missing heartbeats, communication interruptions, and important software faults.

It is essential to fault tolerance because the system must distinguish normal monitoring from monitoring that has become unreliable.

Priority 10: US-06 — Analyze Sediment Plume Movement

Epic: Environmental Analysis

MoSCoW Priority: Must Have

Story Points: 8

This story combines environmental measurements to estimate plume movement and potential boundary crossings.

It has a higher estimate because environmental modeling and interpretation involve significant technical complexity and uncertainty.

Priority 11: US-10 — Initiate Mining Shutdown

Epic: Alert and Shutdown Management

MoSCoW Priority: Must Have

Story Points: 8

This story connects the compliance system to the approved mining control interface and issues shutdown requests when required.

It has a higher estimate because it requires careful safety engineering, control-system integration, acknowledgement handling, and extensive verification.

Priority 12: US-13 — Maintain Compliance Records

Epic: Data Storage and Reporting

MoSCoW Priority: Must Have

Story Points: 3

This story records environmental measurements, alerts, faults, and shutdown events.

These records help demonstrate whether required monitoring and response procedures were followed.

Priority 13: US-15 — Manage User Permissions

Epic: Security and Access Control

MoSCoW Priority: Must Have

Story Points: 5

This story provides role-based access control for environmental data and safety-related functions.

It is important for preventing unauthorized access and protecting sensitive system operations.

Priority 14: US-12 — Recover Data After Communication Failure

Epic: Fault Detection and Recovery

MoSCoW Priority: Should Have

Story Points: 5

This story enables the system to synchronize locally buffered sensor readings after communication is restored.

It improves data recovery and reduces permanent data loss during temporary outages.

Priority 15: US-14 — Generate Compliance Reports

Epic: Data Storage and Reporting

MoSCoW Priority: Should Have

Story Points: 3

This story allows authorized users to generate reports containing environmental measurements, alerts, faults, and shutdown events.

It supports environmental audits and regulatory review.

Backlog prioritization note: The MoSCoW classifications above describe overall product requirements. Marking a story as Should Have does not remove any mandatory regulatory obligation. The Product Owner must ensure that all required compliance and safety functions are completed and validated before the system is used operationally.

5. Plan the Sprint Backlog

5.1 Sprint Planning Overview

The first sprint is planned for two weeks.

The main purpose of Sprint 1 is to develop a working environmental monitoring prototype that registers devices, collects simulated sensor data, validates the incoming measurements, monitors device health, and displays the available information on a dashboard.

The sprint focuses on building the foundation required for later environmental analysis and compliance enforcement.

Selected User Stories for Sprint 1

US-01: Register Monitoring Devices

- Priority: Must Have.
- Story points: 3.
- Expected result: An administrator can register surface buoys and underwater drones.

US-03: Collect Environmental Data

- Priority: Must Have.
- Story points: 5.
- Expected result: The system can receive simulated turbidity, current, acoustic, and depth observations.

US-04: Validate Sensor Readings

- Priority: Must Have.
- Story points: 3.
- Expected result: The system can identify invalid, duplicate, missing, and stale measurements.

US-02: Monitor Device Status

- Priority: Must Have.
- Story points: 3.
- Expected result: The system can identify basic device-health and connectivity problems.

US-05: View Environmental Conditions

- Priority: Must Have.
- Story points: 5.

- Expected result: The dashboard displays the available environmental data and device status.

Total planned effort: 19 story points.

This is a preliminary estimate based on an assumed development team. The team should confirm the amount of work it can complete after considering its actual capacity and technical constraints.

5.2 Reasons for Selecting These User Stories

1. Device registration establishes the identity of every monitored device.
2. Environmental data collection provides the input required for subsequent processing.
3. Data validation helps prevent incorrect measurements from being treated as reliable.
4. Device-health monitoring makes equipment and communication failures visible.
5. Dashboard visualization allows stakeholders to inspect the environmental information.

At the end of Sprint 1, the team should be able to demonstrate a working workflow from simulated sensor data generation to dashboard visualization.

The sprint does not include live mining shutdown control. That feature requires separate safety analysis and verification before operational deployment.

5.3 Sprint Task Breakdown

US-01: Register Monitoring Devices — 3 Story Points

The development tasks are:

1. Identify the required device information.
2. Design the device registration data model.
3. Implement the registration function.
4. Validate device IDs and required fields.
5. Prevent duplicate device registration.
6. Write tests for successful and unsuccessful registration.

Expected output: A working device registration feature.

US-03: Collect Environmental Data — 5 Story Points

The development tasks are:

1. Define the environmental sensor data format.
2. Specify the required device ID and timestamp fields.
3. Build a sensor data simulator for the prototype.
4. Implement the data ingestion component.
5. Support configured turbidity, current, acoustic, and depth observations.
6. Add error handling for unsupported messages.

7. Test the ingestion component using sample data.

Expected output: A working environmental data collection pipeline using simulated sensor readings.

US-04: Validate Sensor Readings — 3 Story Points

The development tasks are:

1. Define data validation rules.
2. Check required fields and data types.
3. Identify duplicate observations.
4. Implement checks for stale or invalid readings.
5. Record validation failures.
6. Test the component using valid and invalid datasets.

Expected output: A validation component that identifies invalid and stale measurements.

US-02: Monitor Device Status — 3 Story Points

The development tasks are:

1. Define device-health status values.
2. Implement a basic device heartbeat mechanism.
3. Configure the heartbeat timeout for the prototype.
4. Update device status when health information is received.
5. Detect missed heartbeats.
6. Test available, unavailable, and unknown status.

Expected output: A basic device-health monitoring feature.

US-05: View Environmental Conditions — 5 Story Points

The development tasks are:

1. Design the initial dashboard layout.
2. Display registered device information.
3. Display the latest validated sensor measurements.
4. Show measurement timestamps.
5. Display device-health status.
6. Identify stale or missing data visually.
7. Test the dashboard using valid and invalid readings.

Expected output: A dashboard displaying environmental measurements and device status.

5.4 Sprint 1 Acceptance Criteria

US-01: Register Monitoring Devices

Given an administrator is logged in, when the administrator enters a unique device ID, device type, and required information, then the system saves the device and displays its registration status.

The system must reject duplicate IDs and missing required fields.

US-03: Collect Environmental Data

Given a registered test device is available, when it sends a supported environmental observation, then the system receives and associates the measurement with its source device and timestamp.

Unsupported data formats must be identified and logged.

US-04: Validate Sensor Readings

Given an environmental measurement has been received, when the validation component checks the required fields and configured quality rules, then the measurement is marked valid or invalid, with a reason recorded for invalid data.

Stale readings must be distinguishable from current readings.

US-02: Monitor Device Status

Given a device is registered, when a valid health update is received or the device misses its configured heartbeat timeout, then the system updates the device status and displays it to an authorized user.

The system must distinguish unavailable or unknown devices from devices confirmed to be operating.

US-05: View Environmental Conditions

Given validated sensor measurements are available, when an authorized operator opens the dashboard, then the dashboard displays the latest readings, device identifiers, timestamps, and device-health status.

The dashboard must clearly identify stale and missing data.

6. Estimate Effort Using Story Points

6.1 Selected Estimation Technique: Planning Poker

Planning Poker is used to estimate the relative complexity and effort of each user story.

During estimation, team members review the user story, discuss its requirements, and independently select a story-point value. The estimates are then discussed until the team reaches an agreed estimate.

The Fibonacci sequence is used:

1, 2, 3, 5, 8, 13, 21

Story points represent relative effort, technical complexity, uncertainty, and testing requirements. They do not directly represent hours or days.

6.2 Story Point Estimation Scale

1 Story Point

A very small task with minimal complexity and uncertainty.

Example: Adding a simple display field.

2 Story Points

A small task with limited complexity.

Example: Validating a basic input field.

3 Story Points

A moderately sized task with a clear implementation approach.

Example: Registering a device and rejecting duplicate IDs.

5 Story Points

A medium-sized task involving multiple components or more extensive testing.

Example: Collecting and validating environmental sensor messages.

8 Story Points

A complex task involving several components, technical uncertainty, or extensive testing.

Example: Developing and validating plume analysis or integrating shutdown requests.

13 Story Points

A very large or highly uncertain story that should generally be divided into smaller stories.

Example: Developing an extensive environmental prediction subsystem.

21 Story Points

An extremely large story that must be divided into smaller requirements before implementation.

Example: Developing a complete end-to-end environmental compliance platform.

6.3 Justification for Sprint 1 Estimates

US-01: Register Monitoring Devices — 3 Points

This story requires a device data model, registration logic, input validation, and duplicate detection.

The requirements are reasonably clear, and integration complexity is limited.

Therefore, 3 story points are assigned.

US-03: Collect Environmental Data — 5 Points

This story requires data format definition, ingestion logic, sensor simulation, error handling, and testing of multiple data types.

Supporting different environmental measurements introduces moderate complexity.

Therefore, 5 story points are assigned.

US-04: Validate Sensor Readings — 3 Points

This story requires checks for required fields, data types, timestamps, duplicate observations, and stale readings.

The work is moderately complex but can be implemented as a separate validation component.

Therefore, 3 story points are assigned.

US-02: Monitor Device Status — 3 Points

This story requires heartbeat processing, timeout detection, status updates, and basic fault handling.

The initial implementation is limited to monitoring device health.

Therefore, 3 story points are assigned.

US-05: View Environmental Conditions — 5 Points

This story requires dashboard development, integration with environmental data, device-status displays, and handling of missing or stale readings.

It combines interface development with data integration and testing.

Therefore, 5 story points are assigned.

6.4 Total Sprint Estimate

The Sprint 1 estimate is calculated as follows:

- US-01: Register Monitoring Devices — 3 points.
- US-03: Collect Environmental Data — 5 points.
- US-04: Validate Sensor Readings — 3 points.
- US-02: Monitor Device Status — 3 points.
- US-05: View Environmental Conditions — 5 points.

Total = 3 + 5 + 3 + 3 + 5 = 19 story points.

Therefore, the planned effort for Sprint 1 is **19 story points**.

This estimate must be reviewed according to the actual team's experience, available capacity, and technical discoveries.

7. Present the Sprint Goal and Expected Deliverables

7.1 Sprint Goal

The goal of Sprint 1 is to develop a basic environmental monitoring prototype that registers surface buoys and underwater drones, collects simulated environmental readings, validates sensor data, detects basic device-health problems, and displays the available measurements on a dashboard.

This sprint establishes the foundation for implementing environmental boundary detection, plume analysis, fault recovery, alerts, and shutdown functions in subsequent sprints.

7.2 Expected Sprint Deliverables

1. Device Registration Module

A working module that registers surface buoys and underwater drones and prevents duplicate device IDs.

2. Environmental Data Collection Module

A prototype component that receives supported environmental measurements from simulated monitoring devices.

3. Data Validation Module

A validation component that identifies invalid, duplicate, missing, or stale sensor information.

4. Device-Health Monitoring Module

A basic mechanism for monitoring device status and detecting missed heartbeats.

5. Environmental Monitoring Dashboard

A dashboard that displays device identifiers, available measurements, timestamps, and device-health status.

6. Testing and Validation Report

A report describing the test cases, test results, identified defects, and remaining issues.

7. Updated Product Backlog

The Product Owner and development team review completed stories, revise remaining priorities where necessary, and prepare for the next sprint.

7.3 Sprint Review

At the end of the sprint, the development team demonstrates the working prototype to the Product Owner and relevant stakeholders.

The demonstration includes:

1. Registration of a surface buoy or underwater drone.
2. Generation of simulated environmental readings.
3. Receipt and validation of sensor measurements.
4. Identification of invalid or stale readings.
5. Detection of a simulated device-health problem.
6. Display of environmental information on the dashboard.

Stakeholders provide feedback, which may lead to changes in the product backlog.

7.4 Sprint Retrospective

The development team reviews the sprint to identify what worked well and what needs improvement.

The discussion focuses on:

1. Whether the user stories were sufficiently clear.
2. Whether the story-point estimates were reasonable.
3. Whether the planned tests were completed.
4. Whether integration between the data collection and dashboard modules caused problems.
5. Whether communication and collaboration among team members could be improved.

The team agrees on specific improvements for the following sprint.

8. Proposed Future Sprint Plan

The project is divided into six planned sprints. Each sprint focuses on a specific development objective and builds on the previous sprint's results.

Sprint 1: Establish Basic Environmental Monitoring

Main Objective: Develop the foundation of the environmental monitoring system.

Planned Features:

1. Device registration.
2. Simulated environmental data collection.
3. Sensor data validation.
4. Basic device-health monitoring.
5. Initial environmental monitoring dashboard.

Expected Outcome: A working prototype that registers devices, receives simulated environmental readings, validates data, and displays monitoring information.

Sprint 2: Implement Compliance Configuration and Environmental Analysis

Main Objective: Develop the initial environmental analysis and compliance configuration features.

Planned Features:

1. Configuration of approved environmental boundaries.
2. Storage of environmental safety thresholds.
3. Basic sediment plume analysis.
4. Processing of turbidity and current measurements.
5. Visualization of environmental conditions and estimated plume movement.

Expected Outcome: A system that can analyze available environmental data and compare it with configured monitoring requirements.

Sprint 3: Implement Violation Detection and Alerts

Main Objective: Develop automated environmental violation detection and alert generation.

Planned Features:

1. Boundary checking.
2. Environmental risk classification.
3. Detection of defined safety violations.
4. Warning and critical alert generation.
5. Environmental event recording.

Expected Outcome: A system that identifies environmental violations, generates appropriate alerts, and records the associated events.

Sprint 4: Develop Fault Tolerance and Recovery

Main Objective: Improve monitoring reliability when devices or communication links fail.

Planned Features:

1. Communication failure detection.
2. Local buffering of environmental data.
3. Synchronization after communication is restored.
4. Device-health monitoring improvements.
5. Redundant monitoring and processing where appropriate.
6. Fail-safe behavior when essential monitoring is unavailable.

Expected Outcome: A monitoring system that can tolerate selected failures and recover from supported communication interruptions.

Sprint 5: Integrate and Verify Safety Enforcement

Main Objective: Integrate the compliance system with the approved mining shutdown interface.

Planned Features:

1. Development of the shutdown request interface.
2. Integration with the mining control system in an approved test environment.
3. Shutdown acknowledgement handling.
4. Verification of shutdown response.
5. Testing of safety boundary violations.
6. Independent review of safety-related behavior.

Expected Outcome: A tested safety-enforcement component that issues shutdown requests for defined violations and records the results.

Live operational deployment must only occur after the required safety verification and approvals are completed.

Sprint 6: Complete Security, Reporting, and System Validation

Main Objective: Complete supporting system features and perform end-to-end validation.

Planned Features:

1. Role-based user access control.
2. Environmental compliance report generation.
3. Audit log management.
4. Security testing.
5. Performance and reliability testing.
6. End-to-end system validation.
7. Documentation and deployment preparation.

Expected Outcome: An integrated environmental compliance system with reporting, access control, fault-handling capabilities, and documented test results.

The sprint sequence is a preliminary plan. The Product Owner should review priorities and dependencies at the end of each sprint. Safety analysis must begin in the first sprint and continue throughout development rather than being postponed until the shutdown integration sprint.

9. Definition of Ready and Definition of Done

9.1 Definition of Ready

A user story is considered ready for a sprint when:

1. The user story is clearly written.
2. Its business value is understood.
3. Acceptance criteria are documented.
4. Dependencies are identified.
5. Required data and interfaces are understood.
6. The story is small enough to complete within a sprint.
7. The development team has estimated the effort.
8. Safety-related risks have been identified where applicable.

9.2 Definition of Done

A user story is considered done when:

1. The functionality has been implemented.
2. The code has been reviewed.
3. Unit tests have passed.

4. Relevant integration tests have passed.
5. Acceptance criteria have been verified.
6. Identified defects have been resolved or formally documented.
7. Relevant documentation has been updated.
8. Security requirements have been checked where applicable.
9. The feature has been integrated into the working prototype.
10. The Product Owner has accepted the story based on the agreed acceptance criteria.

For safety-related functionality, the Definition of Done must also include appropriate hazard analysis, requirements traceability, independent verification, and safety approval.

10. Conclusion

The Agile Sprint Planning and User Story Workshop provides a structured approach to developing the Deep-Sea Mining Environmental Compliance System.

The project vision focuses on reliable environmental monitoring, early detection of sediment plume violations, and prompt safety responses. The requirements have been divided into epics and user stories covering device management, data collection, data validation, environmental visualization, environmental analysis, compliance monitoring, shutdown management, fault recovery, reporting, and security.

Acceptance criteria have been defined using the Given–When–Then format to make each requirement clear and testable. The product backlog has been prioritized according to environmental safety, stakeholder value, technical dependencies, and implementation risk.

For the first two-week sprint, five high-priority user stories have been selected with a preliminary total of 19 story points. The sprint focuses on building a working prototype for device registration, simulated environmental data collection, data validation, basic device-health monitoring, and dashboard visualization.

The Scrum framework supports incremental development, continuous testing, and stakeholder feedback. However, the safety-critical nature of the project requires careful requirements management, systematic fault analysis, independent verification, and appropriate safety reviews before operational deployment.

Overall, this sprint plan establishes a practical foundation for progressively developing and validating a fault-tolerant environmental compliance system for deep-sea mining operations.